

MalmstenIntegralSequences

Package Guide

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Purpose

`MalmstenIntegralSequences` is a Wolfram Language package for the closed forms and finite coefficient systems used in the Malmsten-integral sequence work. The package is deliberately narrow: it keeps the symbolic reductions, coefficient vectors, and nested-sum right hand sides in one loadable place, so the paper and the notebooks can refer to the same computations.

Loading the package

After the paclet is installed, load it in the usual way:

```
Needs["MalmstenIntegralSequences`"]
```

For local work from the repository, the kernel file can be loaded directly:

```
Get["MalmstenIntegralSequences/Kernel/MalmstenIntegralSequences.wl"]
```

Conventions

The sequence index n is a positive integer throughout. The scaled Malmsten integral is used in the positive-parameter range of the paper:

$$\int_0^\infty \log(ax) \operatorname{sech}^n(bx) dx, \quad a > 0, \quad b > 0.$$

The package leaves symbolic parameters symbolic unless a reduction is part of the stated formula. The Hurwitz-zeta difference helpers also handle removable singularities at $s = 1$, avoiding the misleading intermediate form $\infty - \infty$.

Typical session

For nested sums, the most readable route is to keep the right hand side symbolic first and evaluate it only when needed:

```
rhs = NestedFSymbolicRHS[4, {2, 2, 2, 2}]
```

This gives

$$10 \mathcal{F}[2] + 4 \mathcal{F}[3] + \mathcal{F}[4].$$

Then

`EvaluateFRHS[rhs]`

substitutes the reduced values of the F_j terms. The direct command

`NestedFRHS[4, {2, 2, 2, 2}]`

returns the reduced right hand side without the intermediate symbolic display. These two routes are intended to agree.

Main functions

Function	Role
<code>SechPowerIntegral[n]</code>	Closed form for $\int_0^\infty \text{sech}^n x \, dx$.
<code>MalmstenSequence[n,a,b]</code>	Closed form for $\int_0^\infty \log(ax) \text{sech}^n(bx) \, dx$.
<code>ChiSequence[n]</code>	Closed form for the χ_n sequence.
<code>FSequence[n]</code>	The same reduced value, named for the paper's script F_n .
<code>LambdaSequence[n]</code>	Computes λ_n from $n\lambda_n = \chi_n + \lambda_1$.
<code>DeltaSequence[n]</code>	Computes $\delta_n = \chi_{n+1} - \chi_n$.
<code>NestedChiCoefficients[N, lows]</code>	Coefficient vector of $\chi_1, \dots, \chi_N$ in the nested sum.
<code>NestedFRHS[N, lows]</code>	Reduced right hand side using <code>FSequence</code> .
<code>NestedFSymbolicRHS[N, lows]</code>	The same right hand side left in symbolic script- F form.
<code>EvaluateFRHS[expr]</code>	Substitutes symbolic script- F terms and reduces the result.
<code>NestedChiIntegrandPolynomial[N, lows, x]</code>	Polynomial in $\text{sech } x$ for the nested-sum integrand.
<code>ReduceQuarterZetaTerms[expr]</code>	Reduction rules for quarter-argument Hurwitz-zeta derivative terms.

Examples

`MalmstenSequence[1]`

returns

$$\pi \log\left(\frac{\Gamma(3/4)\sqrt{2\pi}}{\Gamma(1/4)}\right).$$

`MalmstenSequence[4]`

returns the reduced even-power value

$$-\frac{2\gamma}{3} - \frac{7\zeta(3)}{3\pi^2} + \frac{2}{3} \log\left(\frac{\pi}{4}\right).$$

`NestedChiCoefficients[5,{2,2,2,2,2}]`

returns

$$\{0, 35, 15, 5, 1\},$$

so the corresponding symbolic right hand side is

$$35F_2 + 15F_3 + 5F_4 + F_5.$$

The optional third argument in `NestedFSymbolicRHS[N, lows, f]` changes only the symbolic head used for display. The two-argument form is the natural input for `EvaluateFRHS[expr]`.

Which command should I use?

- Use `MalmstenSequence` for the logarithmic sech^n integral.
- Use `ChiSequence`, `LambdaSequence`, and `DeltaSequence` for the named sequences in the paper.
- Use `NestedChiCoefficients` when the coefficient vector itself is the object of interest.
- Use `NestedFSymbolicRHS` when writing or checking an identity in a human-readable F_j notation.
- Use `EvaluateFRHS` or `NestedFRHS` when a reduced closed form is needed.
- Use `ReduceQuarterZetaTerms` on intermediate expressions containing quarter-argument Hurwitz-zeta derivative terms.

Tests

The test file is `Tests/KernelTests.wlt`. From the package directory, run

```
wolframscript -file Tests/KernelTests.wlt
```

For a notebook-friendly check with a plain printed verdict, use

```
Get["MalmstenIntegralSequences/Tests/VerifyPackage.wls"]
```

The tests cover package loading, coefficient vectors, representative Malmsten-sequence values, pole and zero cases for the Hurwitz-zeta helpers, and the symbolic F -term evaluation path. A successful run should report a 100 percent success rate, or print a PASS line when using the verification script.